METRIC

MIL-DTL-11751G
15 September 2010
SUPERSEDING
MIL-DTL-11751F
1 February 2000

DETAIL SPECIFICATION

POTASSIUM HYDROXIDE SOLUTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers four classes of aqueous potassium hydroxide solution.

1.2 Classification. The potassium hydroxide solution is of the following classes and unit quantity, as specified (see 6.2):

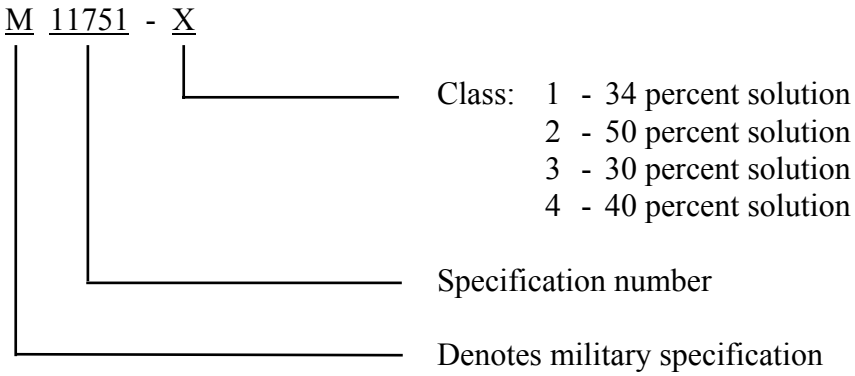
1.2.1 Classes.

Class 1 - 34 percent solution
Class 2 - 50 percent solution
Class 3 - 30 percent solution
Class 4 - 40 percent solution

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1.3 Part or identifying number (PIN). The PIN to be used for potassium hydroxide solution acquired to this specification is created as follows:



2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D891 - Standard Test Methods for Specific Gravity, Apparent, of Liquid Industrial Chemicals

(Copies of this document are available online at http://www.astm.org/ or from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959.)

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA B511 - Potassium Hydroxide

(Copies of this document are available online at http://www.awwa.org/ or from AWWA, 6666 West Quincy Avenue, Denver, CO 80235-3098.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Toxic chemicals, hazardous substances, and ozone depleting substances (ODS). The use of toxic chemicals, hazardous substances, or ODS shall be avoided, whenever feasible.

3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Appearance. Potassium hydroxide solution shall be colorless to pale yellow when tested as specified in 4.5.2.

3.5 Physical and chemical characteristics.

3.5.1 Class 1. Class 1 potassium hydroxide solution shall conform to the chemical characteristics in table I when tested as specified herein.

TABLE I. Class 1 characteristics.

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Potassium hydroxide	32.0	36.0	4.5.3
Potassium carbonate	-	3.0	4.5.3

3.5.2 Class 2. Class 2 potassium hydroxide solution shall conform to the chemical characteristics in table II when tested as specified herein.

TABLE II. Class 2 characteristics.

Characteristic	Minimum	Maximum	Test paragraph
Potassium hydroxide, percent by weight	45.0	50.0	4.5.3
Potassium carbonate, percent by weight	-	3.0	4.5.3
Specific gravity at 15/4°C	1.455	1.515	4.5.4

3.5.3 Class 3. Class 3 potassium hydroxide solution shall conform to the chemical characteristics in table III when tested as specified herein.

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TABLE III. Class 3 characteristics.

Characteristic	Minimum	Maximum	Test paragraph
Potassium hydroxide, percent by weight	29.5	30.5	4.5.3
Potassium carbonate, percent by weight	-	0.90	4.5.3
Chloride	-	to pass test	4.5.5
Nitrogen compounds (as N)	-	to pass test	4.5.6
Phosphate	-	to pass test	4.5.7.3
Sulfate	-	to pass test	4.5.8
Ammonium hydroxide precipitate, percent by weight	-	0.01	4.5.9
Heavy metals (as Ag)	-	to pass test	4.5.10
Iron	-	to pass test	4.5.11
Sodium	-	to pass test	4.5.12.4
Foam	-	to pass test	4.5.13
Particulate	-	to pass test	4.5.14
Filterable non-volatile residue (mg/L) [1]	-	50.0	4.5.15.3

[1] Milligrams per liter (mg/L).

3.5.4 Class 4. Class 4 potassium hydroxide solution shall conform to the chemical characteristics in table IV when tested as specified herein.

TABLE IV. Class 4 characteristics.

Characteristic	Minimum	Maximum	Test paragraph
Potassium hydroxide, percent by weight	39.5	40.5	4.5.3
Potassium carbonate, percent by weight	-	0.90	4.5.3
Chloride	-	to pass test	4.5.5
Nitrogen compounds (as N)	-	to pass test	4.5.6
Phosphate	-	to pass test	4.5.7.3
Sulfate	-	to pass test	4.5.8
Ammonium hydroxide precipitate, percent by weight	-	0.01	4.5.9
Heavy metals (as Ag)	-	to pass test	4.5.10
Iron	-	to pass test	4.5.11
Sodium	-	to pass test	4.5.12.4
Foam	-	to pass test	4.5.13
Particulate	-	to pass test	4.5.14
Filterable non-volatile residue (mg/L)	-	50.0	4.5.15.3

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

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4.2 Inspection conditions. All inspections shall be performed on a lot. A lot shall consist of the potassium hydroxide solution of one class, produced by one manufacturer in no more than 24 hours, at one plant, from the same materials, and under essentially the same manufacturing conditions, provided the operation is continuous. In the event the process is a batch process, each batch shall constitute a lot.

4.3 First article inspection. First article inspection shall be performed on a representative sample of the first lot produced when a first article sample is required (see 3.1). This inspection shall include the requirements in tables I, II, III, and IV, and the tests of 4.5.

4.4 Conformance inspection. Conformance inspection shall be performed in accordance with the inspection provisions set forth herein. The characteristics shown in tables I, II, III, and IV, when tested in accordance with 4.5, shall constitute minimum inspections to be performed by the supplier prior to government acceptance or rejection by lot. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies, submitted to the government for acceptance, comply with all requirements of the contract. Failure of any test, by any sample, shall be cause for rejection of the lot.

4.5 Test methods and procedures.

4.5.1 Test specimen preparation. Containers of potassium hydroxide solution shall be mixed thoroughly by rolling or other means before sampling. A representative specimen of approximately 125 grams (g) shall be removed from each class 1 and class 2 sample container and placed in a suitable, clean, dry container labeled to identify the lot or batch and the container from which it was taken. A representative specimen of approximately 400 milliliters (mL) shall be removed from each class 3 and class 4 sample container and placed in a suitable, clean, clear, dry container labeled to identify the lot or batch and the container from which it was taken. Exposure of the specimen to the atmosphere shall be kept to a minimum to avoid absorption of water and carbon dioxide.

4.5.2 Appearance. Visually examine the specimen for color (see 3.4).

4.5.3 Potassium hydroxide and potassium carbonate. Determine the percentage by weight of potassium hydroxide and potassium carbonate of the specimen in accordance with test methods of AWWA B511.

4.5.4 Specific gravity. Determine the specific gravity of the specimen at 15/4°C in accordance with method A of ASTM D891.

4.5.5 Chloride. Rapidly weigh 100 g of the class 1, class 3, or class 4 specimen or 6 g of the class 2 specimen to within 0.1 g and dilute to 1 liter using carbon dioxide-free water to produce solution "A". Dilute 20 mL of solution "A" with water and filter if necessary through a chloride-free filter. To 5 mL of the diluted solution, add 15 mL of water, 1 mL of nitric acid, and 1 mL of 1.7 percent silver nitrate solution. Any turbidity produced shall be no greater than

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that produced by 0.01 milligrams (mg) of chloride ion in an equal volume of solution containing the quantities of reagents used in the specimen test.

4.5.6 Nitrogen compounds. Dilute 20 mL of solution "A" prepared in 4.5.5 with 50 mL of ammonia-free water in a flask connected through a spray trap to a condenser, the end of which dips beneath the surface of 10 mL of 0.1 *N* hydrochloric acid. Prepare a control by placing 50 mL of ammonia-free water in a similar flask and adding 10 mL of solution "A" and a solution of an ammonium salt containing 0.01 mg of combined nitrogen. To each flask, add 0.5 g of aluminum wire in small pieces, allow to stand for 1 hour, and slowly distill about 35 mL. To each distillate, add 2 mL of freshly boiled 10 percent sodium hydroxide solution, dilute to 50 mL with ammonia-free water, and add 2 mL of Nessler reagent. Any color produced in the specimen solution shall be no greater than that produced in the control.

4.5.7 Phosphate.

4.5.7.1 Ammonium molybdate solution. Dissolve 50 g of ammonium molybdate in 1 liter of 1 *N* sulfuric acid.

4.5.7.2 p-Methylaminophenol sulfate solution. Dissolve 2 g of p-methylaminophenol sulfate in 100 mL of water. To 10 mL of this solution, add 90 mL of water and 20 g of sodium bisulfate. Confirm the suitability of the solution by the following test: Add 1 mL of the solution to each of four solutions containing 25 mL of 0.5 *N* sulfuric acid and 1 mL of the ammonium molybdate solution prepared in 4.5.7.1. Add 0.005 mg of phosphate ion to one of the solutions, 0.01 mg to a second, and 0.02 mg to a third. Allow to stand at room temperature for 2 hours. The three solutions should show readily perceptible differences in blue color corresponding to the relative amounts of phosphate added. The solution to which 0.005 mg of phosphate was added should be perceptibly bluer than the blank control.

4.5.7.3 Procedure. To 40 mL of solution "A" prepared in 4.5.5, add 10 mL of hydrochloric acid and evaporate to dryness on a steam bath. Dissolve the residue in 25 mL of approximately 0.5 *N* sulfuric acid, add 1 mL of the ammonium molybdate solution prepared in 4.5.7.1, and 1 mL of the p-methylaminophenol sulfate solution prepared in 4.5.7.2. Allow to stand for 2 hours at room temperature. Any blue color produced shall be no darker than that produced by 0.02 mg of phosphate ion in an equal volume of solution containing the quantities of reagents used in the specimen test, including the residue from evaporation of 5 mL hydrochloric acid.

4.5.8 Sulfate. Neutralize 250 mL of solution "A" prepared in 4.5.5 with hydrochloric acid and add an excess of 2 mL of hydrochloric acid. Heat to boiling, add 10 mL of 12 percent barium chloride solution, digest in a covered beaker on a steam bath for 2 hours, and allow to stand overnight. If a precipitate is formed, filter, wash thoroughly, and ignite. The weight of the precipitate shall be no more than 0.0012 g greater than the weight obtained in a complete blank test.

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4.5.9 Ammonium hydroxide precipitate. Cautiously add 12 mL of sulfuric acid to 12 mL of water, cool, add the mixture to 100 mL of solution "A" prepared in 4.5.5, and evaporate to dense fumes. Cool, dissolve the residue in 130 mL of hot water, and add ammonium hydroxide until the solution is just alkaline to methyl red. Heat to boiling, filter, wash with hot water, and ignite. Calculate the percent by weight ammonium hydroxide precipitate as follows:

$$\text{Percent ammonium hydroxide precipitate} = \frac{100 \times A}{W}$$

where: A = Weight of residue in grams.
W = Weight of specimen in grams.

4.5.10 Heavy metals. To 30 mL of solution "A" prepared in 4.5.5, cautiously add 15 mL of nitric acid. Prepare a control by adding 0.04 mg of silver ion to 10 mL of solution "A" and cautiously adding 15 mL of nitric acid. Evaporate both solutions to dryness over a low flame or on an electric hot plate. Dissolve each residue in about 20 mL with water. Adjust the pH of each solution to between 3 and 4 (using a pH meter) with 1 *N* acetic acid or ammonium hydroxide solution containing 10 percent ammonia. Dilute to 40 mL with water and mix. Add 10 mL of freshly prepared hydrogen sulfide water to each and mix. Any yellowish brown color produced in the specimen solution shall be no darker than that produced in the control.

4.5.11 Iron. Neutralize 20 mL of solution "A" prepared in 4.5.5 with hydrochloric acid using phenolphthalein indicator. Add 2 mL of hydrochloric acid in excess and dilute to 50 mL with water. Prepare a control by adding 0.01 mg of iron and 2 mL of hydrochloric acid to the residue obtained from evaporation of the quantity of hydrochloric acid used to neutralize the specimen solution. Dilute to 50 mL with water. To each solution, add 30 to 50 mg of ammonium persulfate and 3 mL of 30 percent ammonium thiocyanate solution. Any red color produced in the specimen solution shall be no darker than that produced in the control.

4.5.12 Sodium.

4.5.12.1 Flame photometer. Use a flame photometer that will accurately determine the amount of sodium in the specimen.

4.5.12.2 Control solution. To 10 mL of solution "A", add 0.2 mg of sodium ion and dilute to 100 mL with water.

4.5.12.3 Specimen solution. Dilute 10 mL of solution "A" prepared in 4.5.5 to 100 mL with water.

4.5.12.4 Procedure. Adjust the flame photometer to give the maximum emission with the control solution prepared in 4.5.12.2 at the 589 nanometer (nm) sodium line and note this reading. Without adjusting the flame photometer's controls, determine the emission of the specimen solution prepared in 4.5.12.3 at this wavelength. Change only the monochromator by the amount specified in this test. Observe the emission of the control solution prepared in 4.5.12.2 at 589 nm sodium line. Observe the emission of the specimen solution prepared in

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4.5.12.3 at the 589 nm sodium line and at a wavelength of 580 nm. The difference between the intensities observed for the specimen solution at 580 nm and 589 nm shall not exceed the difference observed at 589 nm between the specimen solution and the control solution.

4.5.13 Foam. A 250 mL quantity of potassium hydroxide solution shall be transferred to a clean, clear, one-half to one-liter container. The container shall be capped and vigorously shaken by hand for 15 seconds, then allowed to stand undisturbed for 5 minutes. There shall be no visible evidence of foam.

4.5.14 Particulate. A 250 mL quantity of potassium hydroxide solution shall be visually examined for clarity and solids in a clean, clear container against a white and black background. The inspection shall be performed by a person with normal visual acuity, natural or corrected, under bright light of not less than 1100 lux (100 foot-candles). A general purpose two-D-cell flashlight with fresh batteries and lamp, positioned 45 centimeters (18 inches) perpendicular to a surface shall illuminate that surface with an intensity of not less than 1100 lux (100 foot-candles). There shall be no visible evidence of particulates, fiber, oil, or grease.

4.5.15 Filterable non-volatile residue (NVR).

4.5.15.1 Balance. The balance shall be accurate to 0.1 mg with a zero drift less than 0.1 mg.

4.5.15.2 Membrane filter. The membrane filter shall be disposable and shall be constructed of pure Teflon® or equal. The membrane filter shall be 47 millimeter in diameter and shall have a pore size of 0.45 or 0.50 micrometer. Bonded or laminated membrane filters shall not be used.

4.5.15.3 Procedure. Weigh a clean, unused, membrane filter to the nearest 0.1 mg. Transfer the membrane filter to a clean filter holder with reservoir. Wet the filter with 1 or 2 drops of isopropyl alcohol. Allow at least 15 seconds for the isopropyl alcohol to fully wet the filter. This is observable as a change in filter color from opaque white when dry to translucent when wet. Filter 10 to 20 mL of deionized water through the filter. Filter 200 mL of potassium hydroxide solution through the filter. Do not exceed a filtration rate of 50 mL per minute. Filter 100 to 110 mL of deionized water through the filter. Do not exceed a filtration rate of 50 mL per minute. Remove the filter, place in a clean dish with cover. Dry the filter with dish cover ajar in an oven at 55 °C to 60 °C (131 °F to 140 °F) for 20 to 30 minutes. Remove the dish with filter from the oven, transfer to an ambient temperature dish and allow it to cool with cover ajar for at least 2 minutes. Weigh the filter to the nearest 0.1 mg. Calculate the potassium hydroxide solution filterable non-volatile residue as follows:

Potassium hydroxide solution filterable NVR = 5 x (B-A)

where: A = Initial weight of the filter in milligrams.
B = Final weight of the filter in milligrams.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

6.1.1 General. Class 1 and class 2 potassium hydroxide solutions are used for the treatment of drinking and industrial water.

6.1.2 Military unique. Class 3 and class 4 potassium hydroxide solutions are used in submarine electrolytic oxygen generators. Use of potassium hydroxide solution contaminated with excessive particulate or organic matter in an electrolytic oxygen generator can lead to a fire or an explosion. Therefore, the purity requirements for class 3 and class 4 potassium hydroxide solutions exceed that of commercially available potassium hydroxide solution.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class of material required (see 1.2).
c. Unit quantity required.
d. Whether a first article is required (see 3.1).
e. Packaging requirements (see 5.1).

6.3 Definition.

6.3.1 Batch. A batch is defined as that quantity of material that has been manufactured by some unit chemical process or subject to some physical mixing operation intended to make the final product substantially uniform.

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6.4 Subject term (key word) listing.

Aqueous
Carbonate
Chemical
Electrolytic
Generator
Oxygen
Percentage
Purification
Submarine
Treatment
Water

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - EA
Air Force - 68
DLA - GS

Preparing Activity:
DLA - GS3

(Project 6810-2010-010)

Review Activity:
Army - MD1

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